Introduction to the BIOMASS Data Innovation and Science Cluster (DISC)

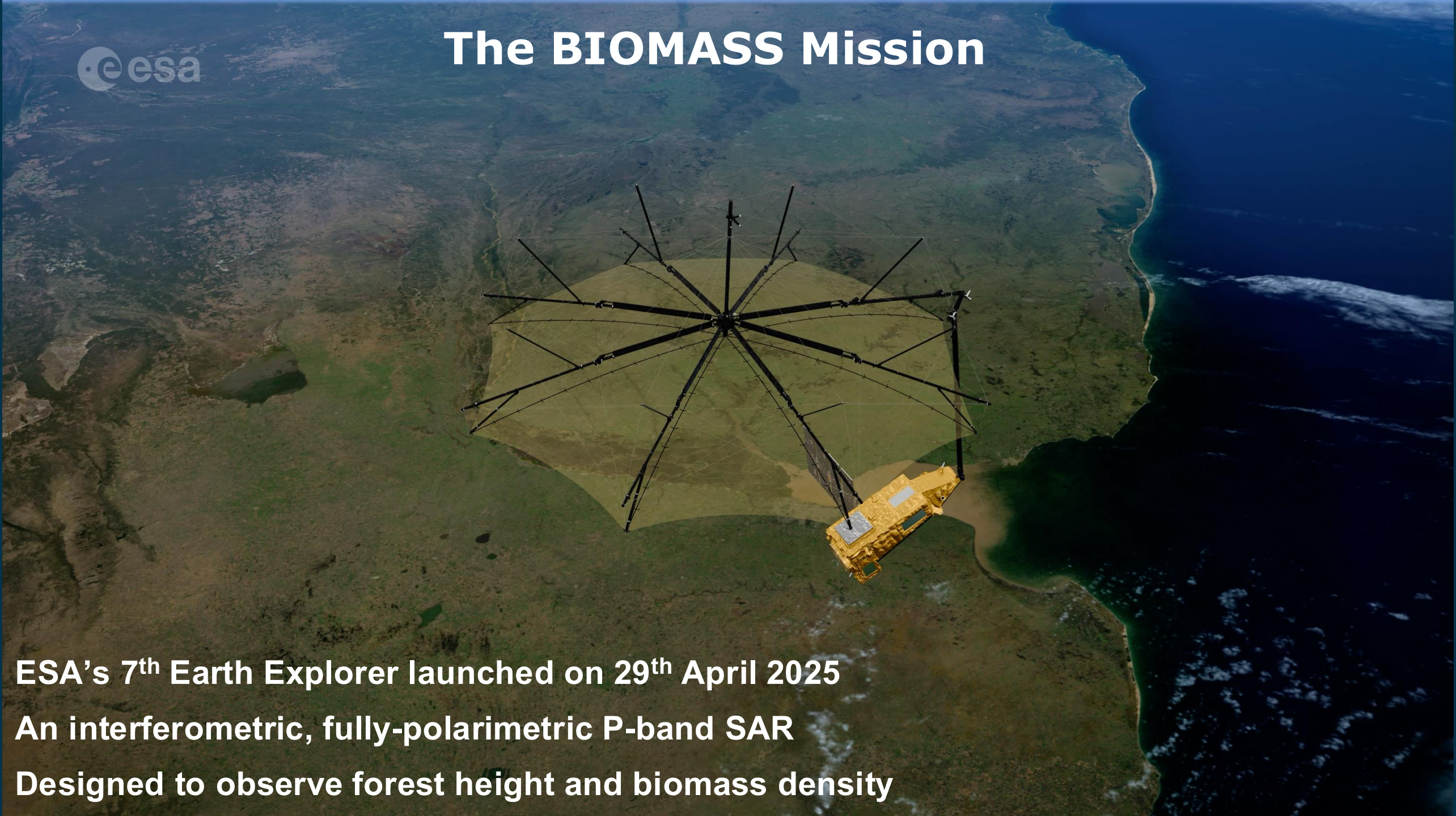
Clément Albinet

Mission Management and Ground Segment Department
European Space Agency

3rd December 2025

ESA UNCLASSIFIED - For ESA Official Use Only

The BIOMASS Mission

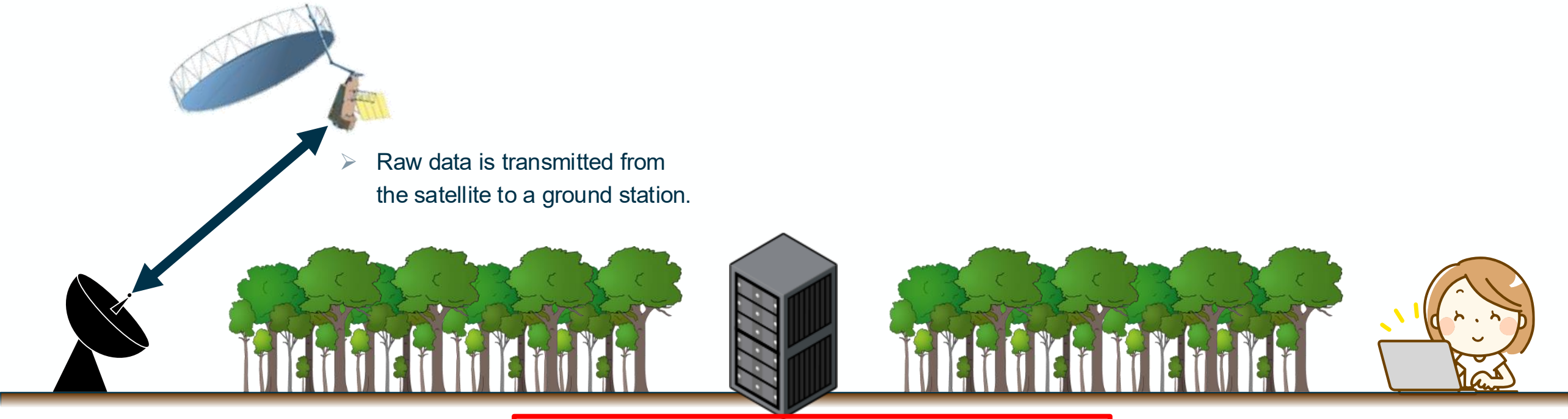


ESA's 7th Earth Explorer launched on 29th April 2025

An interferometric, fully-polarimetric P-band SAR

Designed to observe forest height and biomass density

What is a Payload Data Ground Segment? (summary)



➢ Raw data is transmitted from the satellite to a ground station.

- Data reception: Receives raw instrument data from the satellite.

→

- Data processing: Converts raw data into various product levels (e.g., level 0, level 1, level 2), which involves calibration, correction, geolocation, etc.
- Quality control: Monitors the instruments and performs quality checks on the generated products.
- Archiving: Stores the data and products for long-term use and potential reprocessing.
- Mission Planning: Planning of the instrument acquisitions and downlink station activities.

→

- Data access: Distributes the final, processed data products to users.
- User Services: Supports the user in accessing the data and replies to questions.

DISC

Role of the DISC?

DISC: Data Innovation and Science Cluster

- BIOMASS DISC implements the thematic layer of the ground segment related to:
  - *Quality Control*
  - *Processor maintenance and evolution*
  - *Image formation system (instrument + Processor) calibration*
  - *Forest (L2/3) Product Validation*
  - *Core L2/3 and experimental products generation*

- The DISC is the PDGS front-end towards the scientific community by supporting request and issues troubleshooting (everything beyond data access).

- It is part of the PDGS and interface with FOS (and PLS) in providing expertise on the data for supporting the user community, the instrument anomaly or during normal evolution loop of the overall system.

Highlights

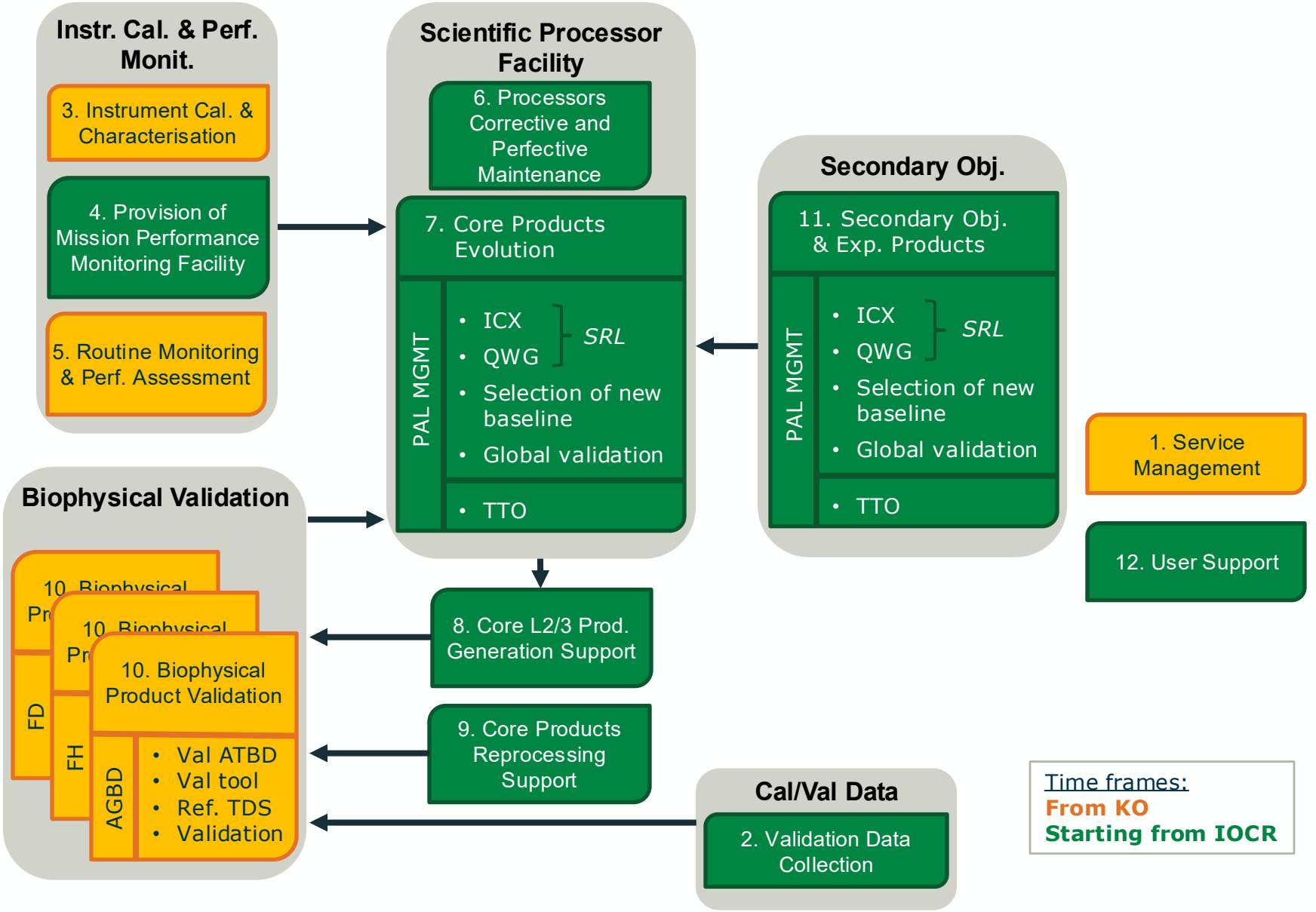
- BIOMASS DISC involves routine process related to the PDGS:
  - Routine operations : day-to-day offline verification of the imaging system performance
  - Routine Calibration process based on the heritage from phase E1 (transponder operations, etc)
  - Routine maintenance of the L1 and L2/3 processors

- A unique entry point for all maters on data scientific content.

- BIOMASS DISC includes a scientific context on the:
  - Innovation on the calibration of spaceborne P-band SAR systems.
  - BIOMASS retrieval algorithm.
  - DISC concept relies on expertise of people. It is a limited group with the risk to be perceived as a close group. The challenges are to introduce flexibility on order to embrace new people, ideas, approach and to work in collaboration with other teams with similar problematics (GEDI, NISAR, ALOS, SAOCOM, etc).

- BIOMASS DISC works on a collaborative environment sharing resources (IT, tools, colocation database) and knowledge (algorithm, source code) and data (in-situ, others)

DISC functions



DISC Organisation

Service Manager – E. Jeansou (ACRI-ST)

Contract Manager – A. Riguccini (ACRI-ST)
QA Manager – E. Dardouri (ACRI-ST)
Security Manager – S. Clouaire (ACRI-ST)
Web portal Manager – J-M. Rivet (ACRI-ST)

BioPAL / Open Science Office

Open Science Manager & Community Manager – Y. Rey-Ricord (ACRI-ST)

Scientific coordination

PIs' leader – S. Tebaldini (PoliMi)
ESLs coordination - D. Giudici (ARESYS)

Core Products algorithm leader - F. Banda (ARESYS)
Secondary products algorithm leader - S. Falco (ARESYS)

Instrument Calibration & Performance Monitoring
PI – A. Barros Cardoso da Silva (DLR)

Core Products

- **Level-1** *PI - S. Tebaldini (PoliMi)*
- **AGB** *PI - L. Ulander (Chalmers)*
- **FH** *PI - K. Papathanassiou (DLR)*
- **FD** *PI – L. Ferro-Famil (CESBIO)*

Experimental Products & Secondary Objectives

- **DTM** *PI - S. Tebaldini (PoliMi)*
- **Tomography** *PI - M. Pardini (DLR)*
- **Ionosphere** *PI - S. Tebaldini (PoliMi)*
- **Desert** *PI – M. Zribi (CESBIO)*
- **Ice** *PI - J. Dall (DTU)*

Validation data provision & Geophysical validation
PI – J. Chave (CNRS)

ESL Network

Routine monitoring & perfo assessment

Operations - V. Taurino (ACRI-ST)
SAR expert - A. Recchia (ARESYS)

Processors & tools maintenance

BIOMASS Processing Suite (BPS) and BTK software - R. Piantanida (ARESYS)
STA software manager - G. Amoroso (ARESYS)

MPMF service S[&]T

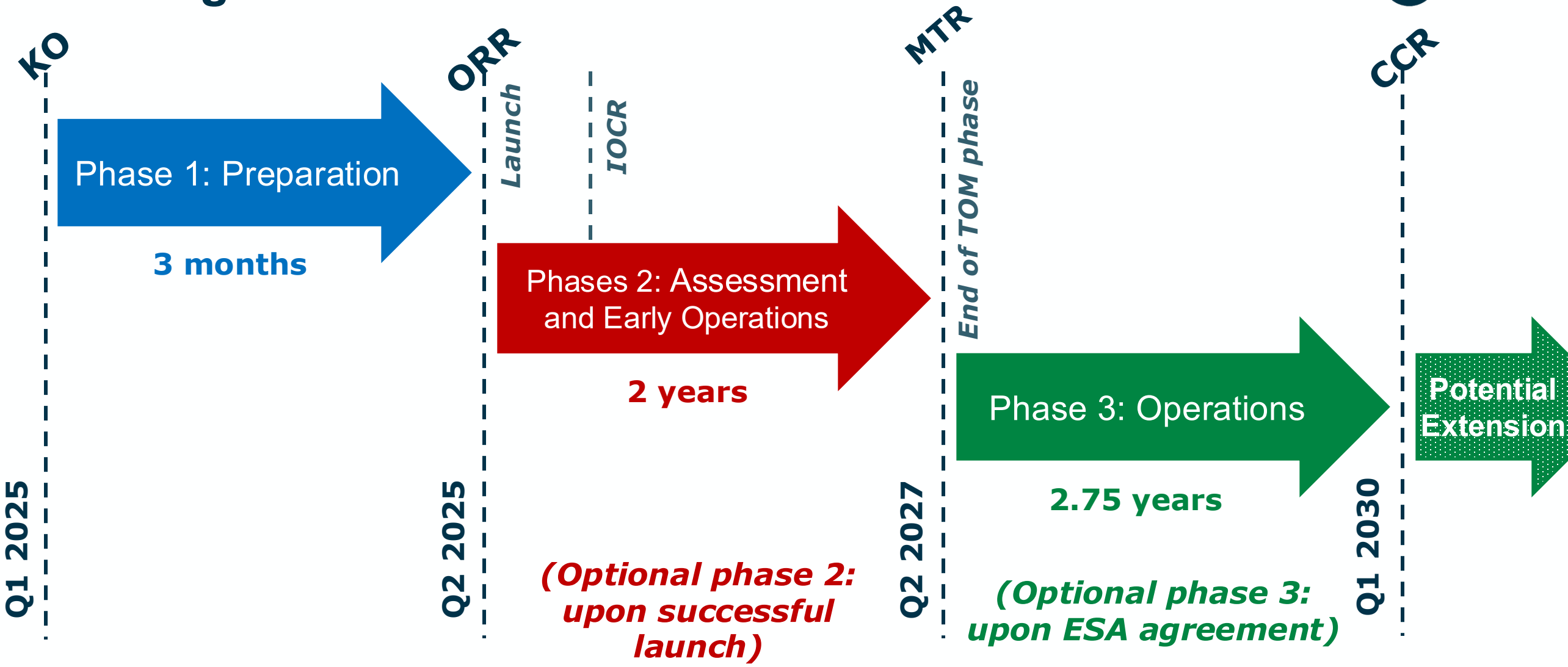
Tech. manager - S.V.Niemeijer
Service leader - R.A. Klavers

Data management support

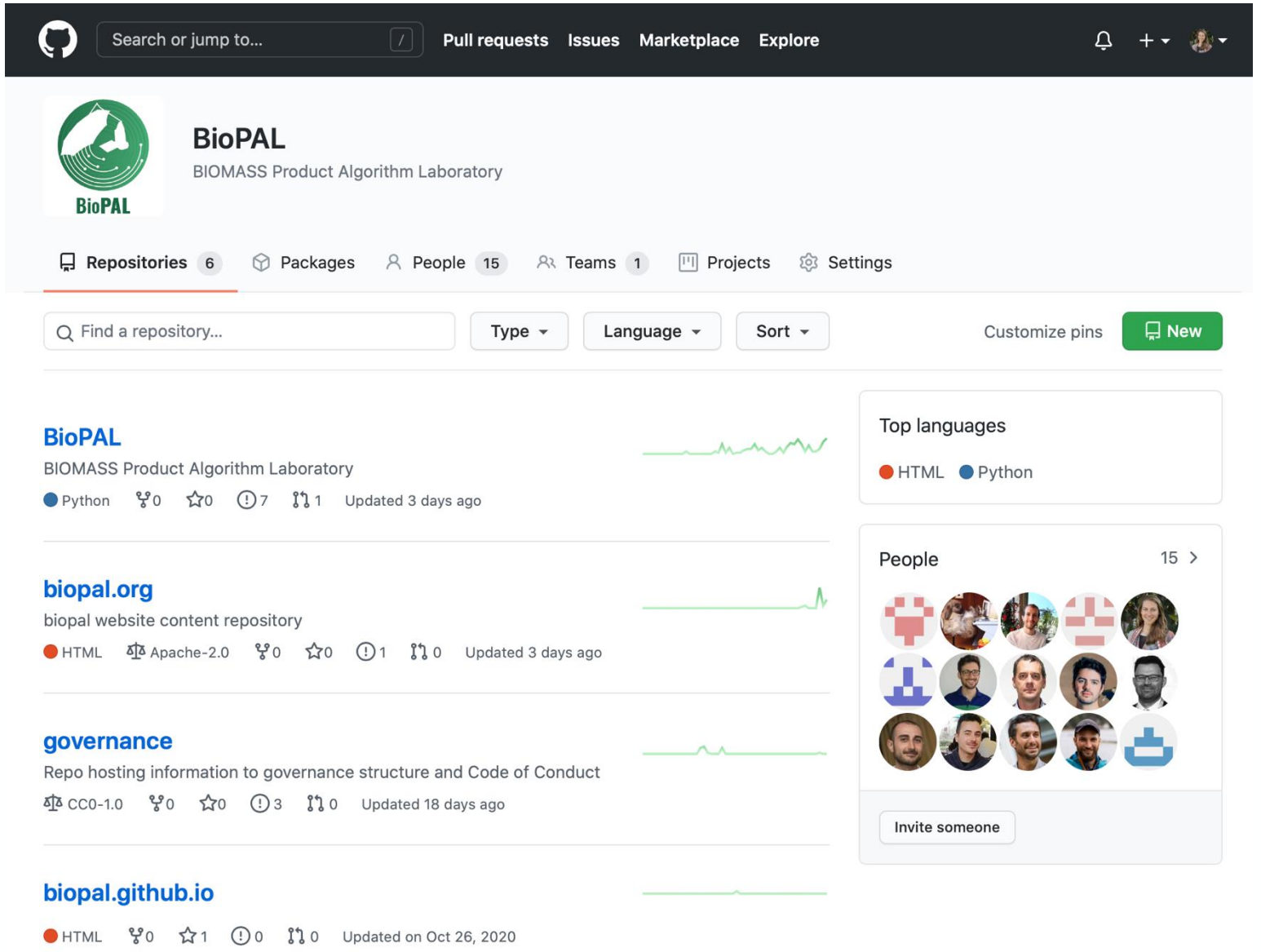
Core L2B/L3 products generation - L. Ferro-Famil (CESBIO)
Core Product reprocessing – F. Rouffi (ACRI-ST)

Technical Office

Phasing


KO
ORR
Launch
IOCR
MTR
End of TOM phase
CCR
Phase 1: Preparation
3 months
Phases 2: Assessment and Early Operations
2 years
Phase 3: Operations
2.75 years
Potential Extension
Q1 2025
Q2 2025
Q2 2027
Q1 2030
(Optional phase 2: upon successful launch)
(Optional phase 3: upon ESA agreement)

New: 1.Open-source Processors



New concept at ESA!

Open source

Today:
Level-2 prototype algorithms

Tomorrow:
Level-1 (as much as possible), Level-2 and Level-3 operational algorithms, other open-source algorithms (Tomo, DTM, etc.)

biopal@esa.int
biopal.org
github.com/BioPAL

New: 2.Open forest biomass reference sites

GEO-TREES consists in a collection of permanent, well-studied, and locally supported sites to support the Cal/Val of the BIOMASS (but also GEDI and NISAR) missions

Sites for validation must be extensive, and cover a range of forest types: tropical, temperate and boreal.

Each site should have:

1. At least 10 1-ha permanent forest plots – established to the highest forest monitoring standards
2. At least 1000 ha of Airborne Lidar Scanning coverage
3. The capacity for Terrestrial Lidar Scanning
4. The capacity to remeasure plots
5. Ancillary data (weather)

- *Following the CEOS WGCV LPV Above Ground Biomass Validation Good Practices*

New: 3.“Mission Algorithm and Analysis Platform”

Baseline for future ESA missions!

→ ***It’s a Virtual open and collaborative environment that…***

Enables researchers to easily discover, process, visualize, and analyze large volumes of data.

Provides tools and infrastructures to bring data into the same coordinate reference frame to enable comparison, analysis, data evaluation, and data generation.

Provides a version-controlled science algorithm development environment that supports tools, co-located data, and processing resources.

Addresses intellectual property and sharing issues related to collaborative algorithm development and sharing of data and algorithms.

New: Product Algorithm Laboratory

New approach!

Algorithm initial definition → First algorithm implementation →

Algorithm improvement by MAAP users:

1- modify processing algorithm/code

2- generate the corresponding new dataset

3- Self Validation by MAAP users

using validation data available in MAAP

4- Verification and approval of new official dataset

Mission Algorithm and Analysis Platform (MAAP)

- *Allow to arrive faster to stable algorithms for R&D missions on a user cooperative approach*
- *People outside the core science team can contribute to the product improvement cycle*

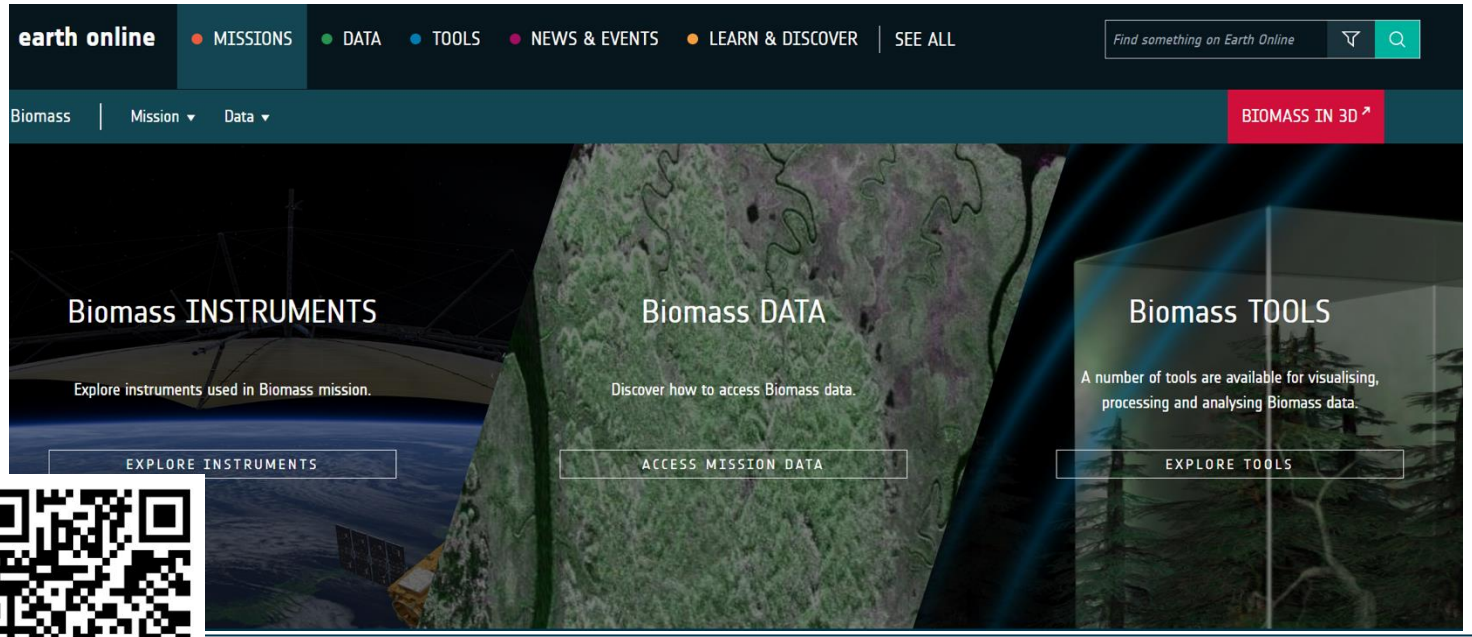
Concepts of "Open Science" → *Well adapted to R&D EO missions*

After Commissioning phase: data and services

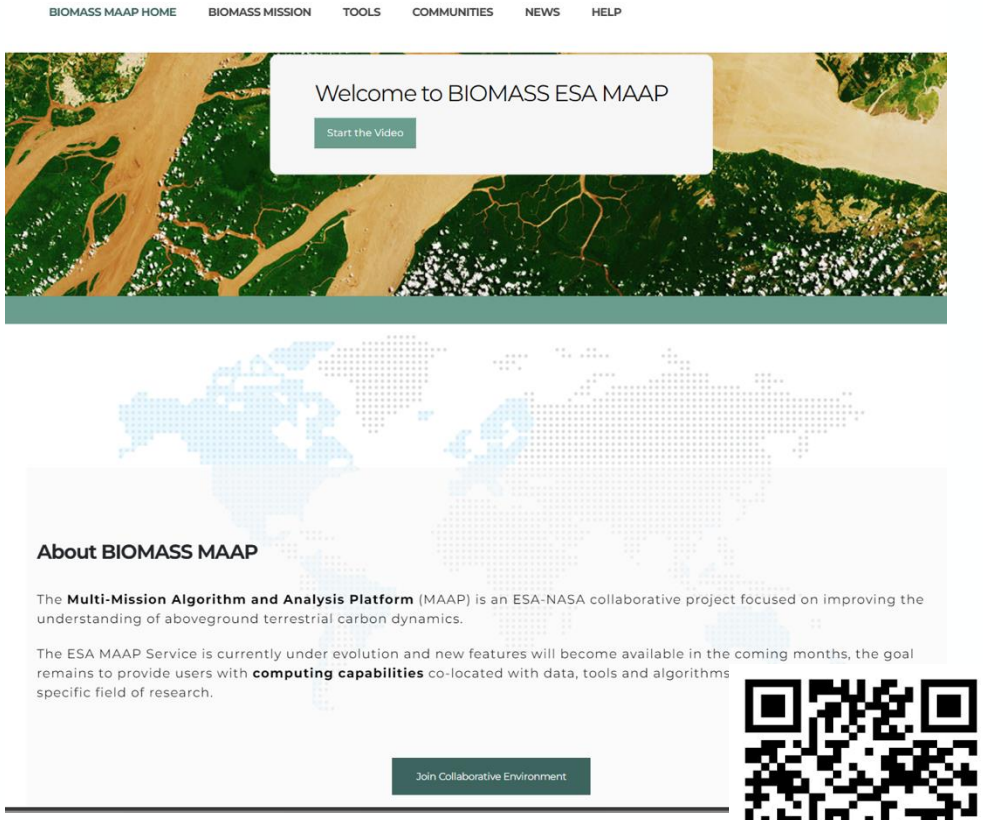
BIOMASS Data and MAAP will progressively be open to users after successful Commissioning phase:

- Target date 12th December 2025

https://earth.esa.int/eogateway/missions/biomass

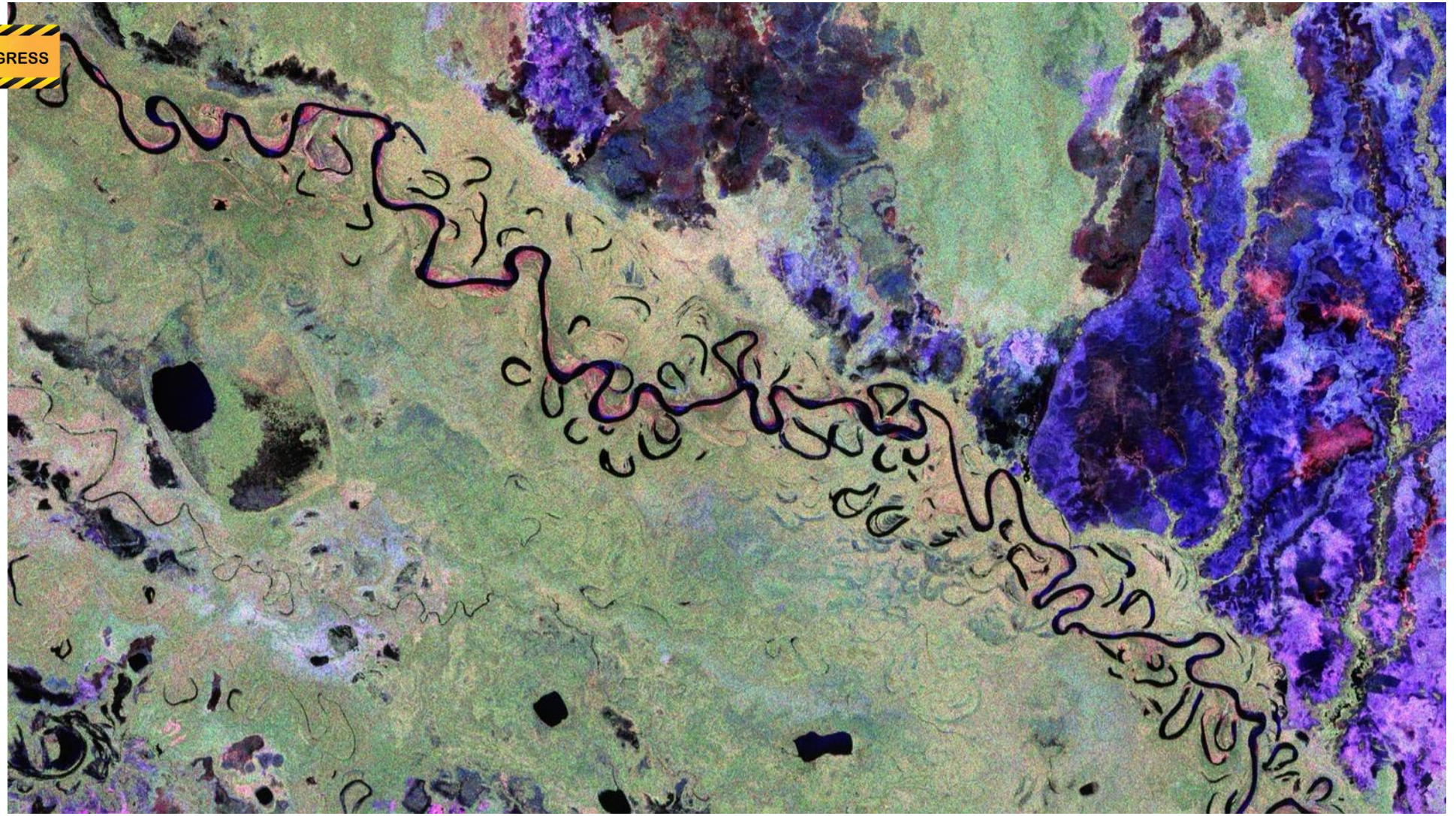


https://portal.maap.eo.esa.int/biomass/



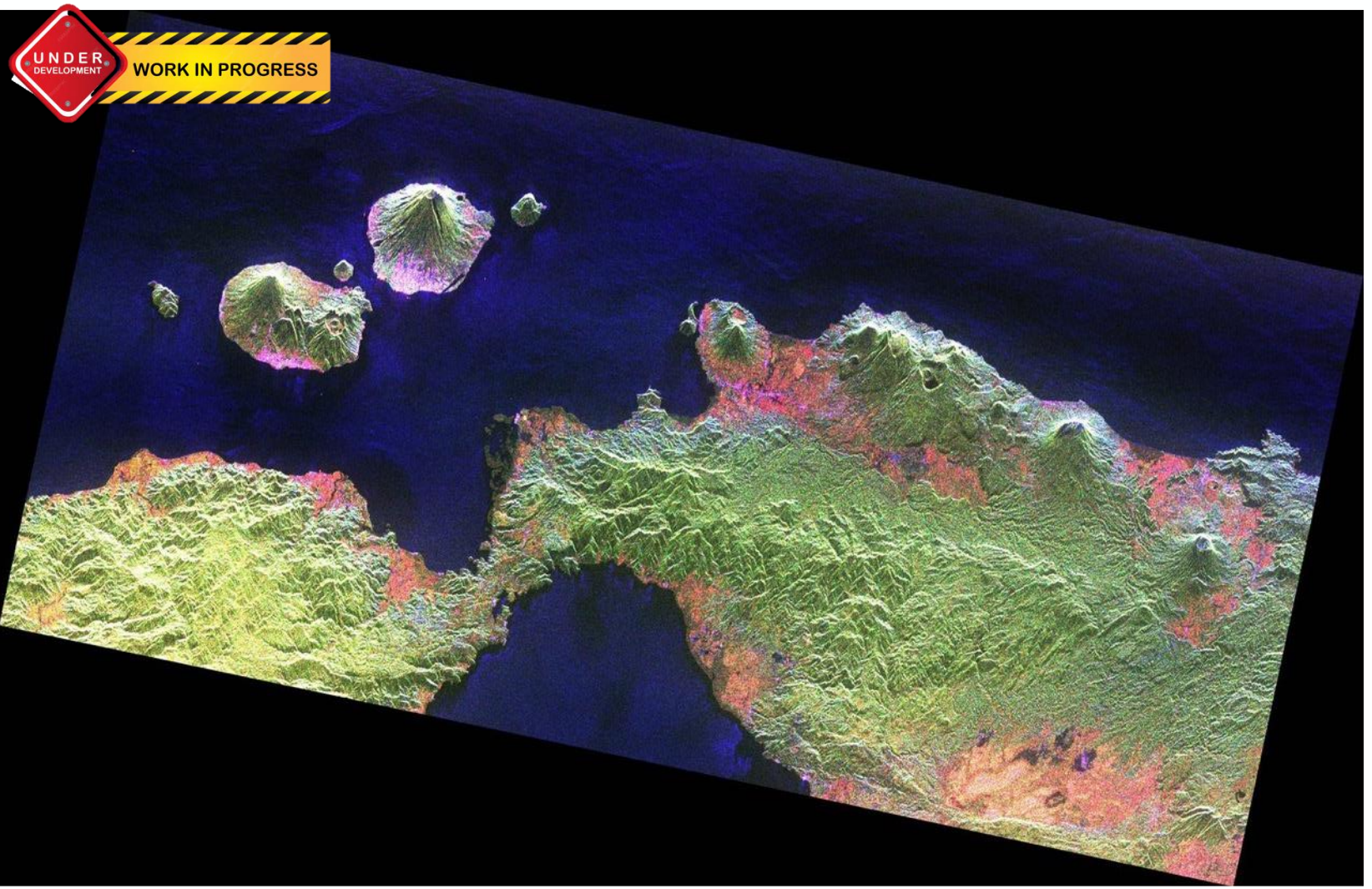
Mission status: First stunning images!

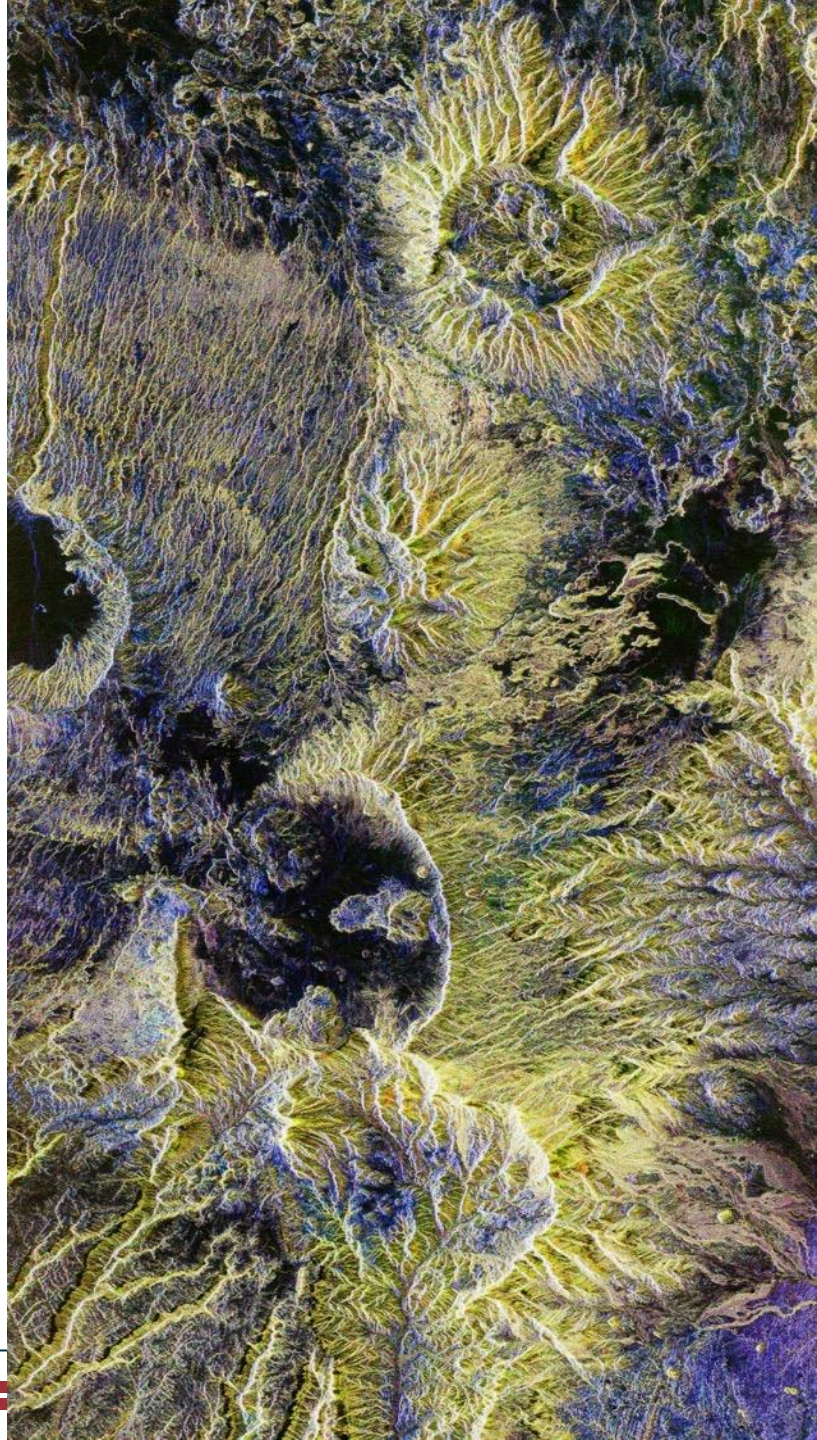
UNDER DEVELOPMENT
WORK IN PROGRESS



Bolivian forest and landscape from Biomass

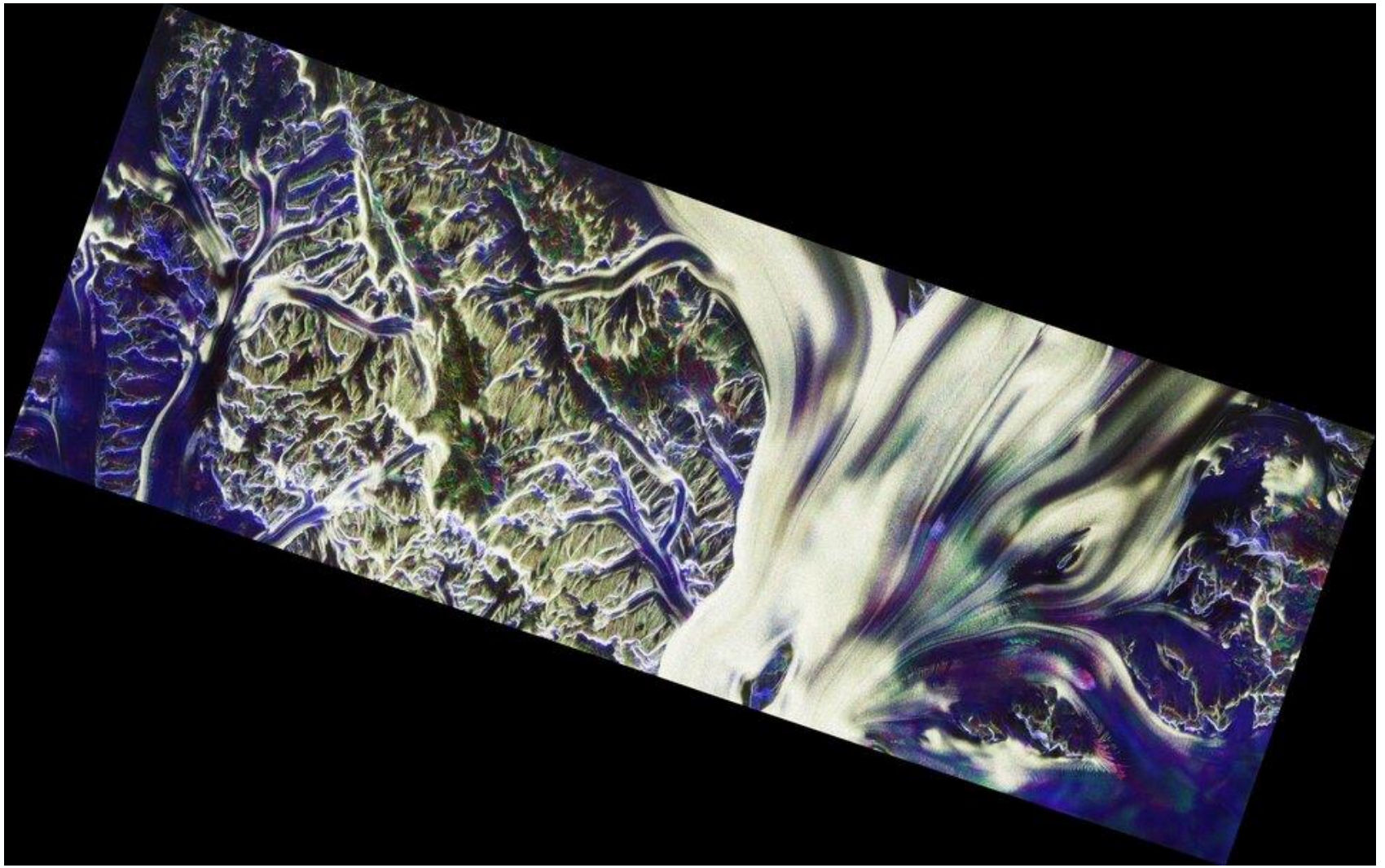
Mission status: First stunning images!





Indonesia – volcanoes on the radar / Desert structure in Chad

Mission status: First stunning images!



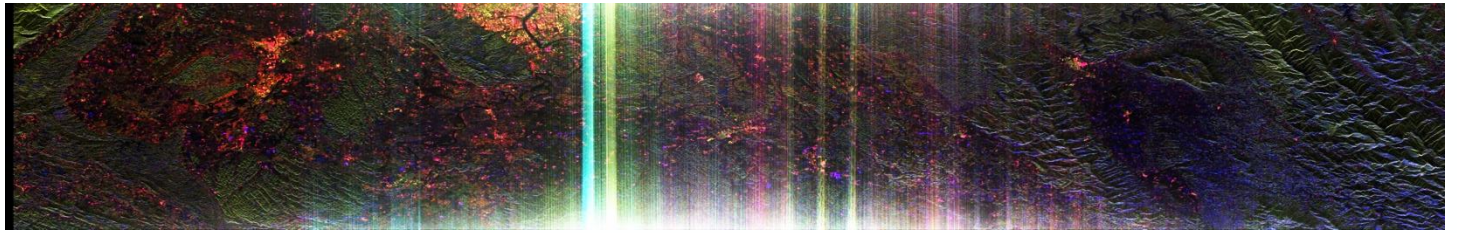
Nimrod Glacier and the Transantarctic Mountains

Several issues still to be addressed

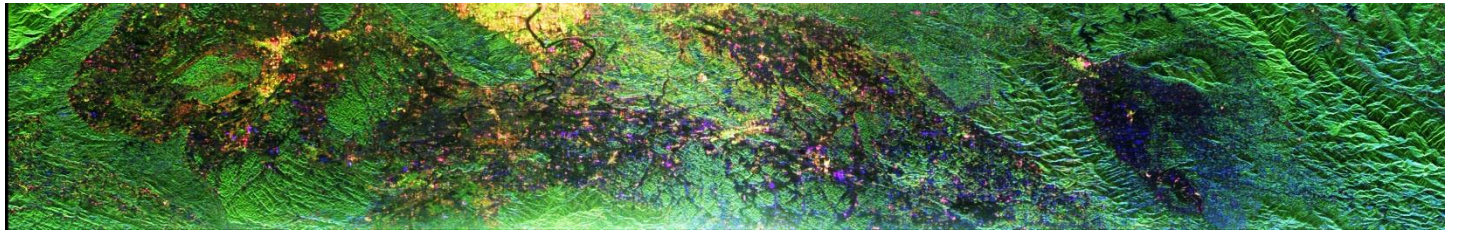
UNDER DEVELOPMENT WORK IN PROGRESS

RFI characterization and mitigation

Heavily affected dataset

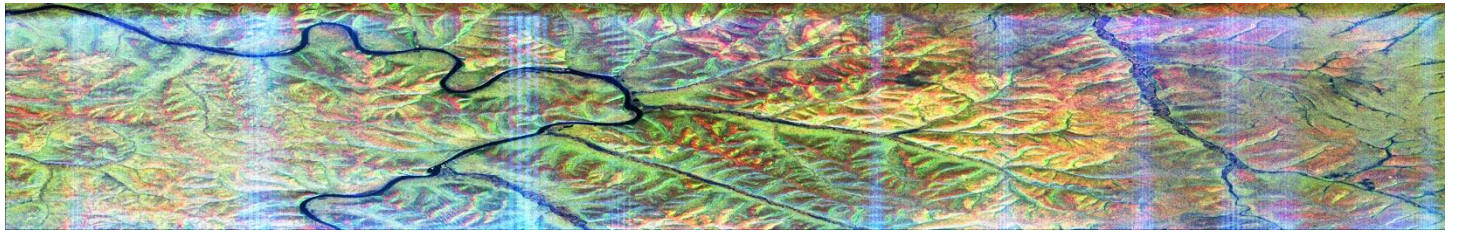


Not mitigated

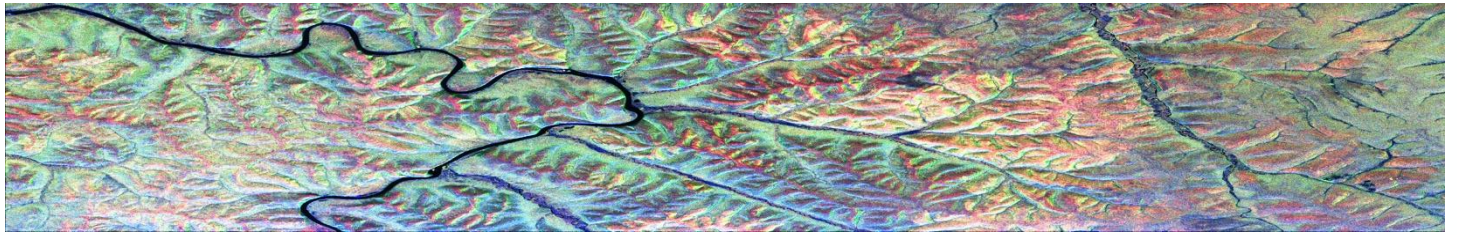


RFI mitigated

Moderately affected dataset



Not mitigated



RFI mitigated

Summary – BIOMASS a true Earth Explorer

1. BIOMASS was proposed in 2005. Implementation started in Nov. 2015. The satellite was launched on 29th April 2025, and the first images processed are extremely promising and showing a great potential. First data for the public will be available on 12th December (target).

2. BIOMASS is the first P-band SAR and first systematic radar tomographic space mission; it is a true Earth Explorer with a lot of unknowns and exciting science for global biomass mapping.

3. The new unique vision of Earth from BIOMASS will extend beyond forests and into measurements of ice, sub-surface geomorphology in deserts, topography, the ionosphere, ocean...

4. It is the first Open-Source Earth Explorer with BioPAL (algorithms), GEO-TREES (in-situ data) and MAAP (platform).